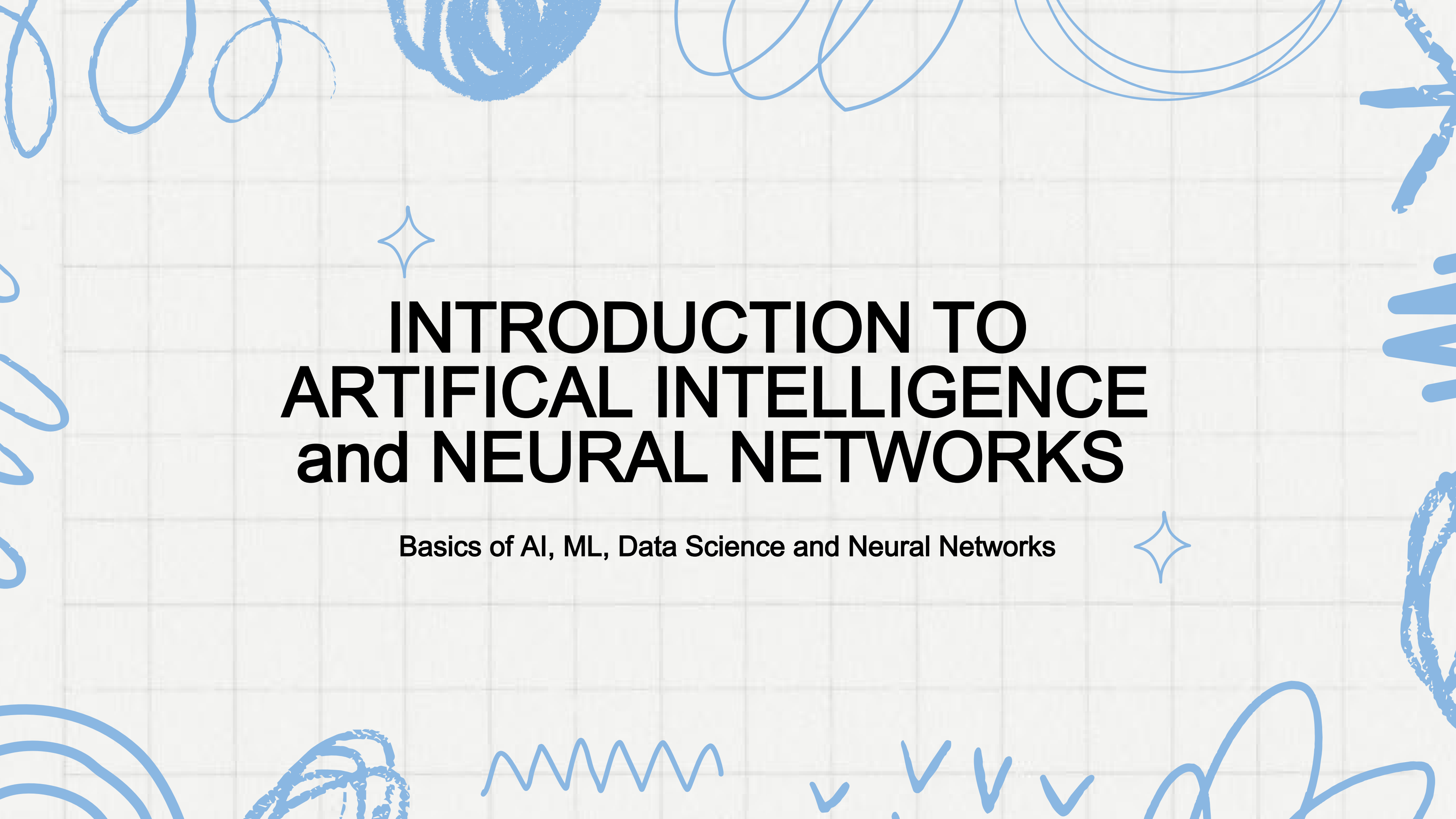


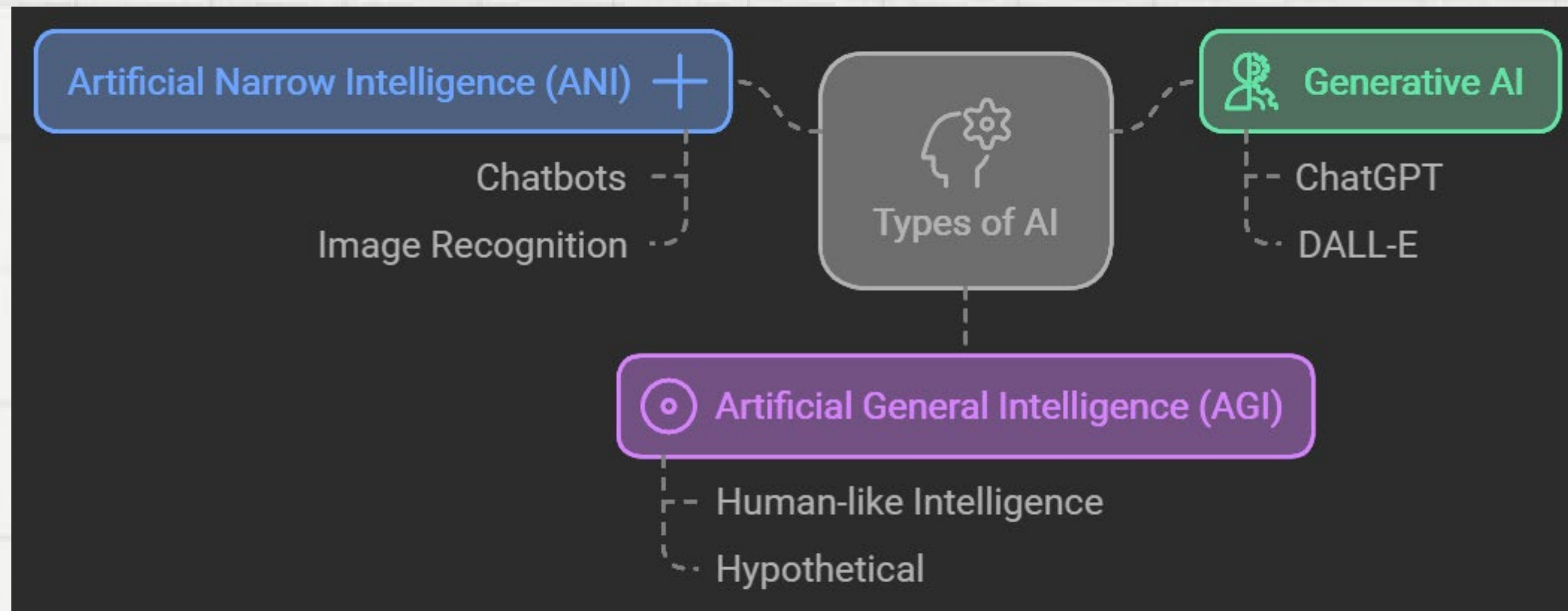


INTRODUCTION TO ARTIFICIAL INTELLIGENCE and NEURAL NETWORKS

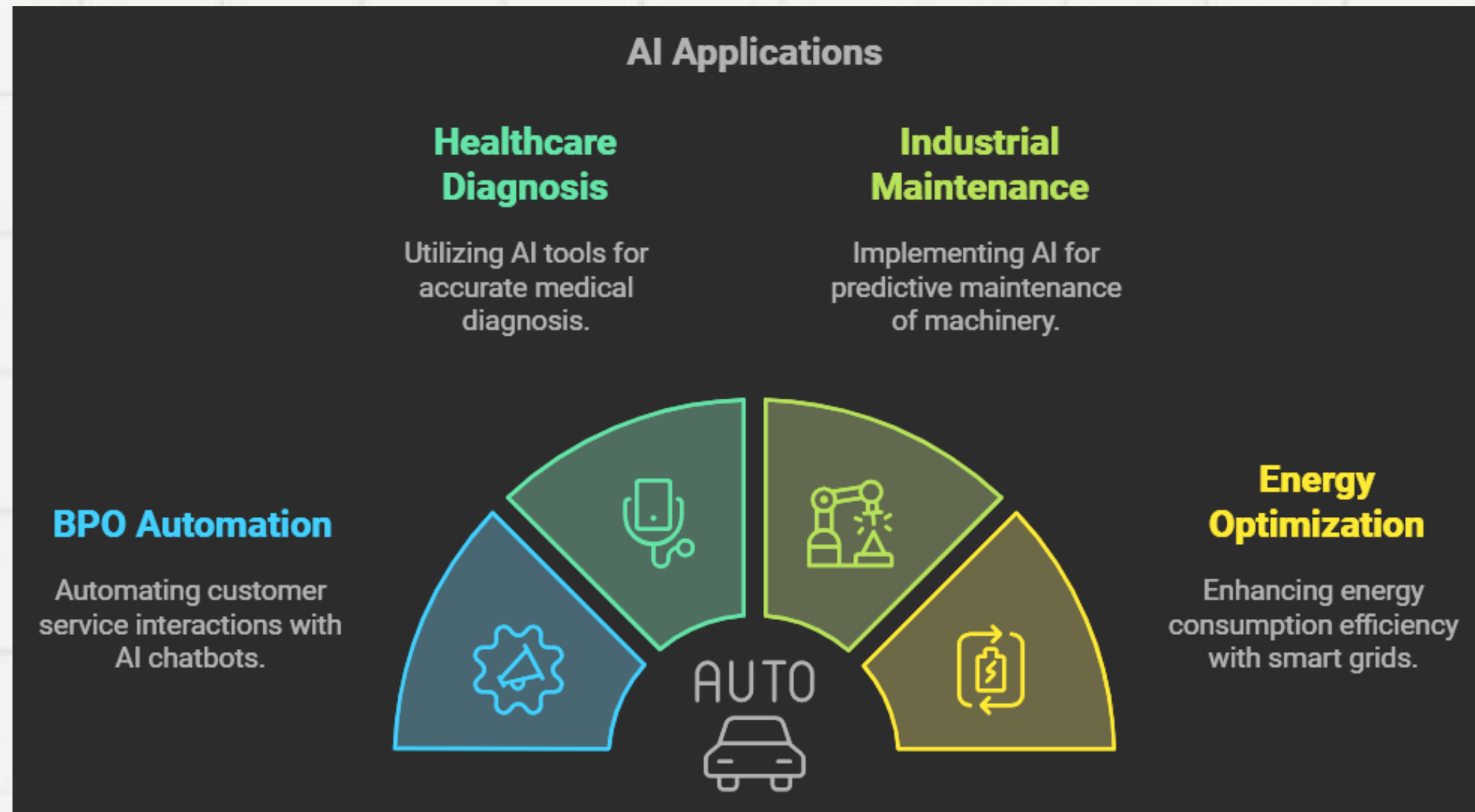
Basics of AI, ML, Data Science and Neural Networks



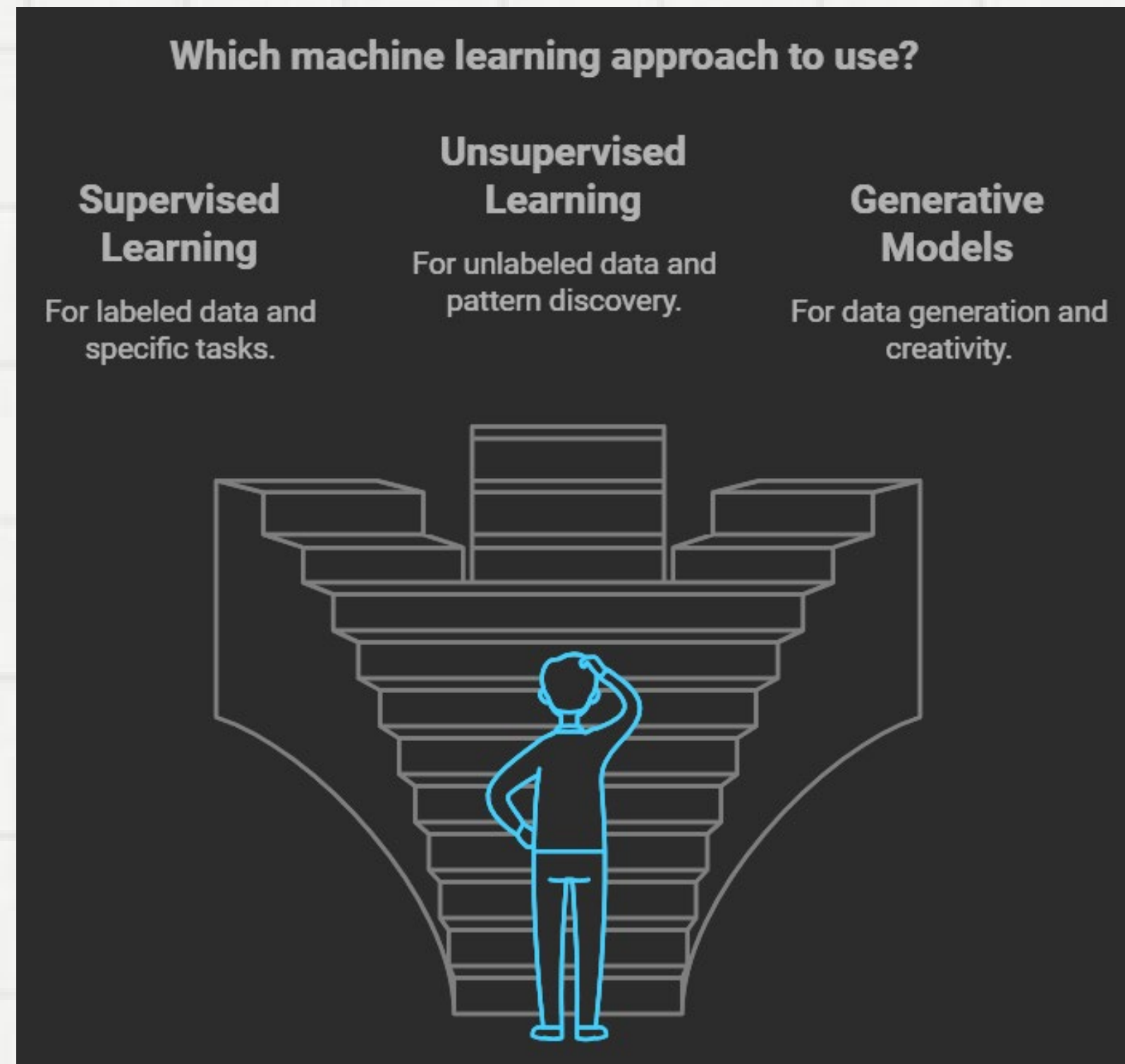
TYPES OF AI

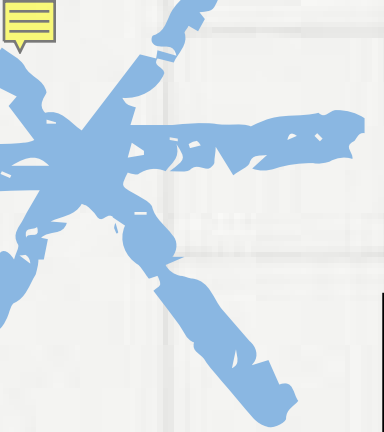


Applications of Artificial Intelligence



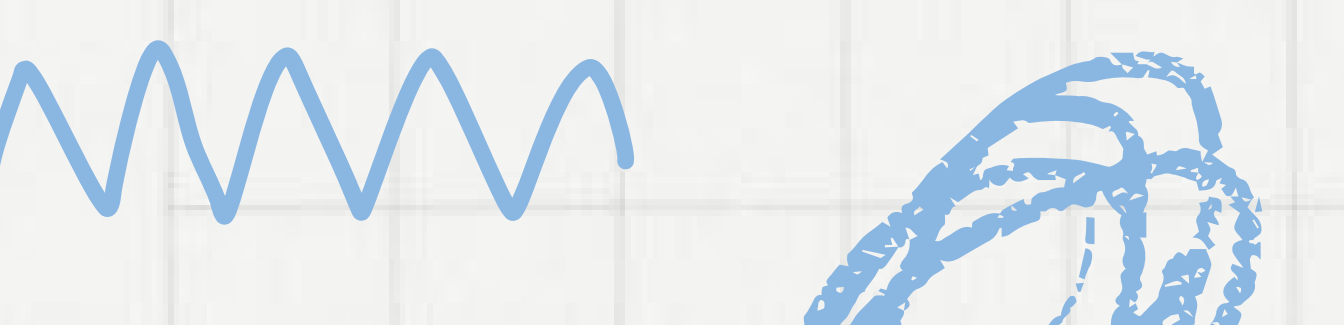
Types of Machine Learning





Differences

FEATURE	DATA SCIENCE	ARTIFICIAL INTELLIGENCE	MACHINE LEARNING
FOCUS	Extracting insights from data	Creating intelligent systems	Making machines learn from data
APPLICATION	Data analysis, visualization	Automation, decision making -	Predictive models
RELATION	Uses AI/ML for insights	Uses ML for decision making -	A subset of AI

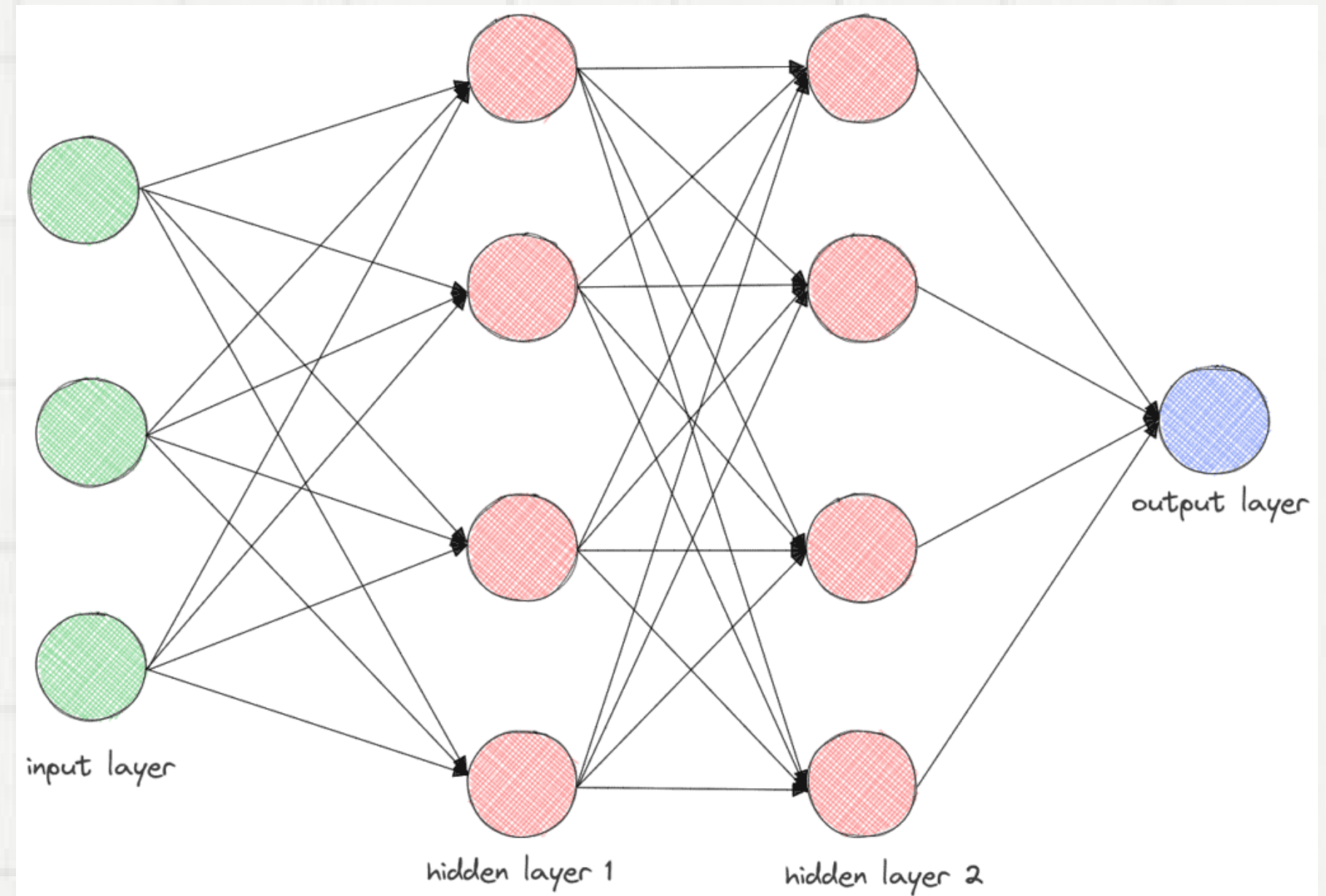


INTRODUCTION TO NEURAL NETWORKS

Neural networks are computational models inspired by the human brain's structure, consisting of interconnected nodes (neurons).

Components:

- Input layer - Represents the input data
- Hidden layer - Shows one or more hidden layers where computations happen.
- Output layer - Displays the final output



Types of Neural Networks

1 Convolutional neural networks (**CNNs**) - input images, identify the objects in a picture, and differentiate them from one another.

pattern recognition, image recognition, and object detection

three main layers - convolutional layer, where most of the computation occurs. pooling layer, where the number of parameters in the input is reduced, the fully connected layer classifies the features extracted from the previous layers.

2 Recurrent neural networks (**RNNs**) - translate language, speech recognition, natural language processing, and image captioning.

smart home technologies – google home, alexa and voice command features on mobile phones - siri

Feedback loops in the structure of RNNs allow information to be stored similarly to how your memory works.

3 Radial basis function (**RBF**) networks: input layer performs no computations. passes the data directly to the hidden layer. faster learning speed.

time series prediction and function approximation

4 Long short-term memory (**LSTM**) networks: unique, sort data into short-term and long-term memory cells (data needs to be looped back into the network as data points or entire sequences)

handwriting recognition and video-to-text conversion

5 Multilayer perceptrons (**MLPs**) : capable of learning the relationship between linear and non-linear data, backpropagation - can reduce error rates.

face recognition and computer vision

6 Generative adversarial networks (**GANs**): generate **new** data sets based on training data

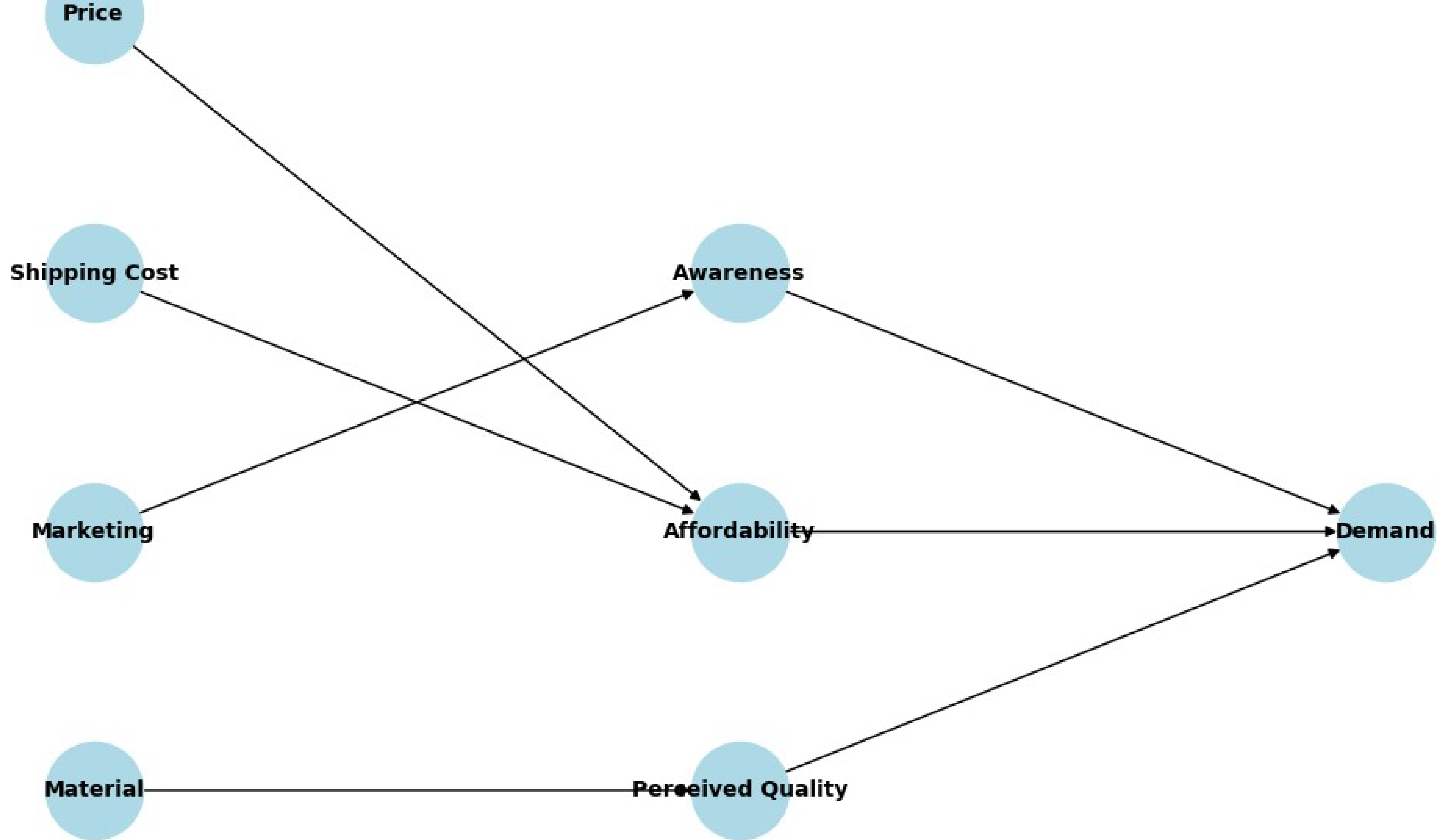
art created with AI, deep fake videos

7 Deep belief networks (**DBNs**): unique, stack individual networks that can use each other's hidden network layers as the input for the next layer, allows for the neural networks to be trained faster.

generate images and motion-capture data. Self driving cars

8 Self-organising maps (**SOMs**), or Kohonen maps: transform extensive complex data sets into understandable two-dimensional maps where geometric relationships can be visualized.

displaying voting trends for analysis and organizing complex data collected by astronomers so it can be interpreted.



Generative AI - Use of Neural Networks

Generative Adversarial Networks (GANs): LMMs (Large Multi Modal language models) use GANs for generating realistic images, videos, and audio.

Variational Autoencoders (VAEs): VAEs are used for generating data by learning the underlying distribution of the input data. They're effective in generating slightly more controlled and interpretable data, often used in image synthesis and anomaly detection.

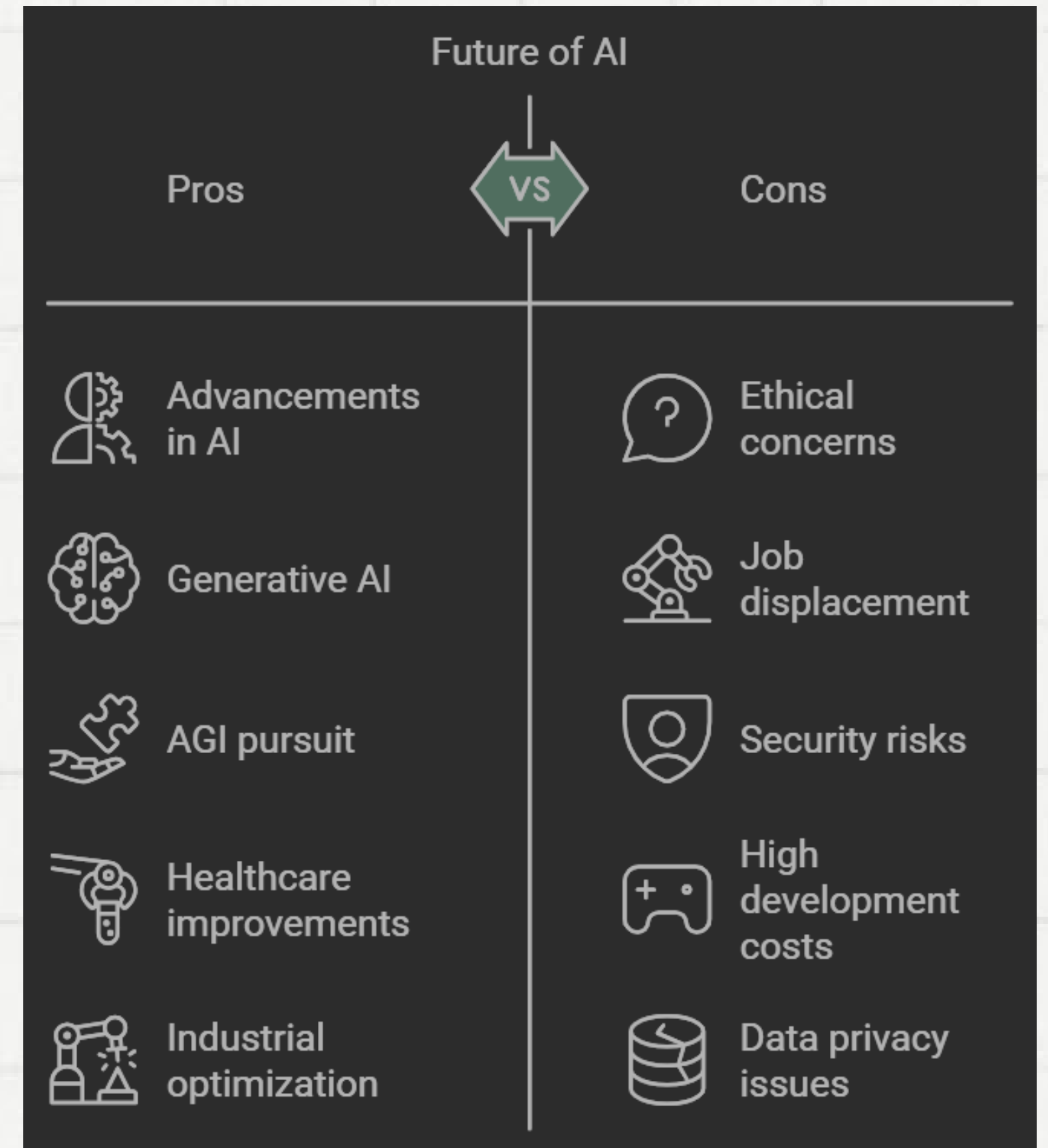
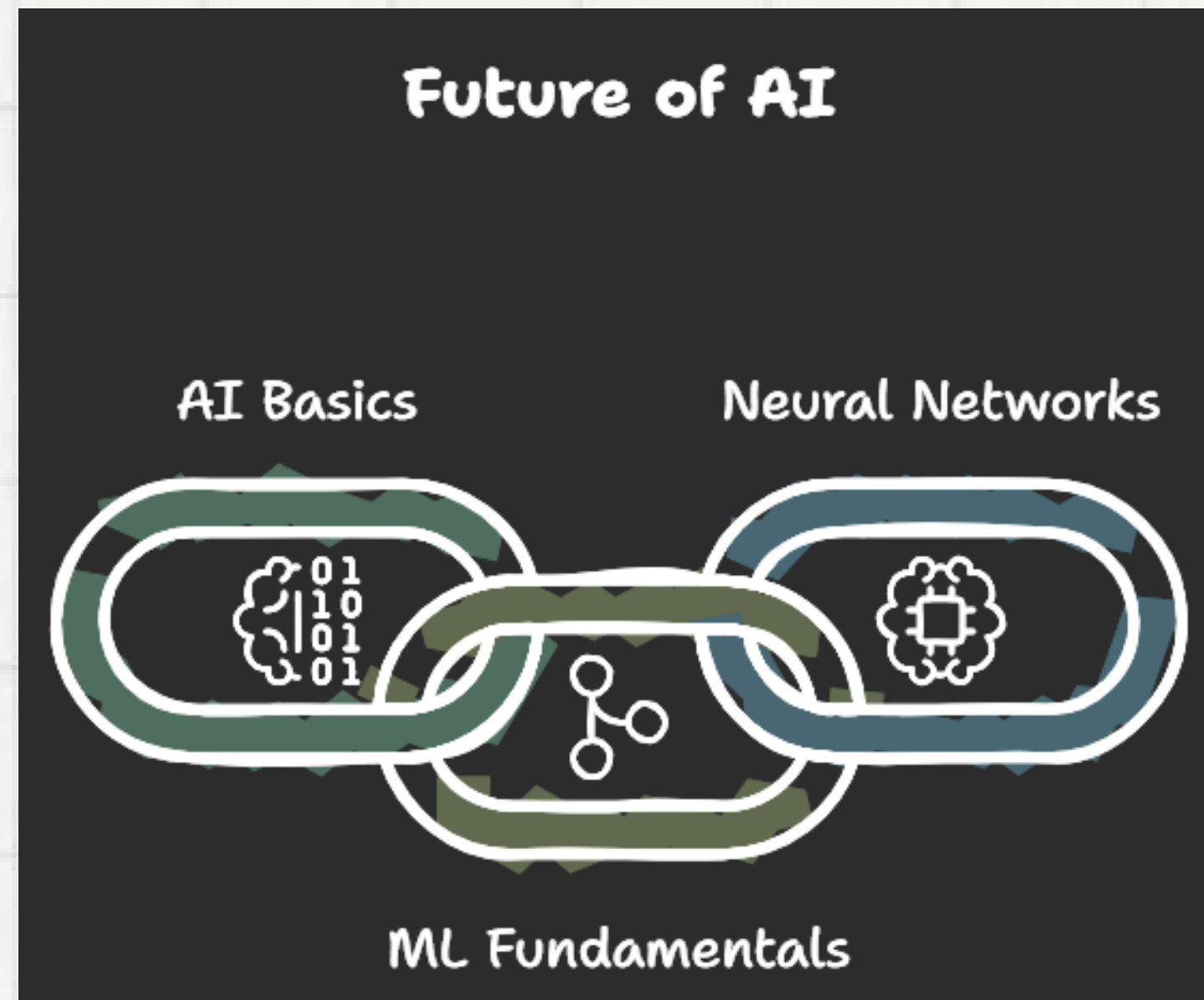
Transformers: large language models (LLMs) like GPT, are crucial in generating text and language-based content. Transformers rely on attention mechanisms rather than recurrence, allowing them to handle longer sequences and generate coherent text, code, or dialogue.

Diffusion Models: popular for generating high-quality images. "denoise" random noise to create clear images, as seen in tools like DALL-E and Stable Diffusion. They're known for creating photorealistic and high-resolution images.

Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) Networks:

Not recently used for Gen AI, RNNs and LSTMs were earlier used in generative text applications, especially where maintaining sequential information was key. They can still be useful in specific sequence generation tasks.

CONCLUSION



The background is a light blue grid. It is decorated with various hand-drawn blue doodles. At the top left, there are several overlapping loops. To the right of these is a solid blue circle. Further right are more overlapping loops. On the far right edge, there is a star-like shape and some horizontal lines. At the bottom left, there are concentric arcs. In the bottom center, there is a wavy line. To the right of the wavy line are several small 'v' shapes. At the bottom right, there is a large, loose loop.

**Thank you
very much!**